Question 3: Steel Sector Production Optimization

1. Problem Description & Setup

The problem involves optimizing a steel manufacturing company's operations and expansion strategy over two periods (2025 and 2035) to minimize total costs while meeting demand and considering potential environmental constraints. The company uses three potential processes (1, 2, and 3), each with different costs, initial capacities, expansion limits, and environmental impacts (water, electricity, fuel consumption, and pollutant emissions).

- Demand: 150 kt/year in 2025, increasing to 200 kt/year in 2035.
- Objective: Minimize the sum of operational and investment costs.

2. Mathematical Model

- **Decision Variables**:
  - x25p: Production level (kt) of process p in 2025 ($p \in \{1,2,3\}$).
  - x35p: Production level (kt) of process p in 2035 ($p \in \{1,2,3\}$).
  - Ep: Capacity expansion (kt) for process p between 2025 and 2035 ($p \in \{1,2,3\}$).

- **Parameters** (from Table 11):
  - cap0: Initial capacity (kt) in 2025 = {1: 120, 2: 50, 3: 0}.
  - delta: Max expansion (kt) by 2035 = {1: 100, 2: 100, 3: 80}.
  - op_cost: Operational cost ($/kt) = {1: 300, 2: 400, 3: 200}.

- inv_cost: Investment cost ($/kt/yr expansion) = {1: 3000, 2: 4000, 3: 5000}.
- water: Water use (units/kt) = {1: 5, 2: 3, 3: 1}.
- electricity: Electricity use (units/kt) = {1: 3, 2: 4, 3: 10}.
- fuel: Fuel use (units/kt) = {1: 6, 2: 8, 3: 3}.
- pollutant: Pollutant emission (units/kt) = {1: 10, 2: 2, 3: 5}.
- demand2025 = 150 kt.
- demand2035 = 200 kt.

- **Objective Function: Minimize Z**:

  $Z = \sum op_cost[p] \cdot x25p + \sum (op_cost[p] \cdot x35p + inv_cost[p] \cdot Ep)$

- **Constraints:**

1. 2025 Capacity: $x25p \leq cap0[p]$ for $p \in \{1,2,3\}$.
2. 2035 Capacity: $x35p \leq cap0[p] + Ep$ for $p \in \{1,2,3\}$.
3. Expansion Limit: $Ep \leq delta[p]$ for $p \in \{1,2,3\}$.
4. 2025 Demand: $\sum = 13x25p \geq demand2025$.
5. 2035 Demand: $\sum = 13x35p \geq demand2035$.
6. Environmental Limits (for Parts b, c, d): Constraints added later based on 2025 baseline levels.
7. Non-negativity: $x25p, x35p, Ep \geq 0$.

3. Code:

```
# steel_nexus.py
# Solve the steel-making LP

import gurobipy as gp
from gurobipy import GRB

# 1. Data (from Table 11)
processes    = [1, 2, 3]
cap0         = {1: 120, 2: 50,  3: 0}    # Installed capacity in 2025 (kt)
delta        = {1: 100, 2: 100, 3: 80}   # Max expansion to 2035 (kt)
op_cost      = {1: 300, 2: 400, 3: 200}  # $ per kt operation
inv_cost     = {1: 3000, 2: 4000, 3: 5000}  # $ per kt-yr of new capacity
# Environmental use/emission
water        = {1: 5,   2: 3,   3: 1}
electricity  = {1: 3,   2: 4,   3: 10}
fuel         = {1: 6,   2: 8,   3: 3}
pollutant    = {1: 10,  2: 2,   3: 5}

demand2025 = 150   # kt
demand2035 = 200   # kt

# --- Calculated Targets for Part b/c (Update if your base case was different) ---
# Baseline 2025: Water=690, Elec=480, Fuel=960, Pollutant=1260
water_target_2035 = 690 * 0.80      # 552
electricity_target_2035 = 480 * 0.80  # 384
fuel_target_2035 = 960 * 0.80      # 768
pollutant_target_2035 = 1260 * 0.80 # 1008
# --- End Targets ---

# 2. Model
m = gp.Model("Steel_Nexus")

# 3. Decision variables
x25 = m.addVars(processes, name="x2025", lb=0)        # production in 2025
x35 = m.addVars(processes, name="x2035", lb=0)        # production in 2035
```

```
E   = m.addVars(processes, name="Expand", lb=0)      # added capacity

# 4. Objective: min ∑ o_p * x25_p + ∑ (o_p * x35_p + i_p * E_p)
m.setObjective(
    gp.quicksum(op_cost[p] * x25[p] for p in processes)
  + gp.quicksum(op_cost[p] * x35[p] + inv_cost[p] * E[p] for p in processes),
    GRB.MINIMIZE
)

# 5. Constraints
cap25_constrs = {}
# 5A. Capacity limits
for p in processes:
    # 2025: production ≤ existing capacity
    cap25_constrs[p] = m.addConstr(x25[p] <= cap0[p], name=f"Cap25_p{p}")
    # 2035: production ≤ existing + expansion
    m.addConstr(x35[p] <= cap0[p] + E[p], name=f"Cap35_p{p}")
    # expansion cannot exceed max delta
    m.addConstr(E[p] <= delta[p], name=f"MaxExp_p{p}")

# 5B. Demand satisfaction
m.addConstr(x25.sum() >= demand2025, name="Demand25")
m.addConstr(x35.sum() >= demand2035, name="Demand35")

# 5C. Environmental Constraints (Uncomment ONE AT A TIME for Part b & c)
# ---> Make sure to assign the active constraint to its variable <---

#water_limit_constr = m.addConstr(gp.quicksum(water[p] * x35[p] for p in processes) <=
water_target_2035, name="WaterLimit35")
#electricity_limit_constr = m.addConstr(gp.quicksum(electricity[p] * x35[p] for p in
processes) <= electricity_target_2035, name="ElectricityLimit35")
#fuel_limit_constr = m.addConstr(gp.quicksum(fuel[p] * x35[p] for p in processes) <=
fuel_target_2035, name="FuelLimit35")
#pollutant_limit_constr = m.addConstr(gp.quicksum(pollutant[p] * x35[p] for p in
processes) <= pollutant_target_2035, name="PollutantLimit35")


# 6. Solve
m.optimize()

# 7. Display results
if m.status == GRB.OPTIMAL:
    print("\nOptimal total cost: $%.2f\n" % m.objVal)
    print("2025 production by process:")
    for p in processes:
        print(f"  Process {p}: {x25[p].x:.2f} kt  (cap={cap0[p]})")
        # Shadow price of capacity constraint in 2025
        # Only print if non-zero (constraint is binding)
```

```
        dual = cap25_constrs[p].Pi
        if abs(dual) > 1e-6:
            print(f"  → Shadow price: ${dual:.2f} per kt")
        else:
            print("") # Keep alignment consistent

    print("\n2035 production by process and expansions:")
    for p in processes:
        cap35 = cap0[p] + E[p].x
        print(f"  Process {p}: produce {x35[p].x:.2f} kt  /  capacity {cap35:.2f} kt
(expanded {E[p].x:.2f} kt)")

    # --- ADDED CODE FOR PART C ---
    # Print shadow prices for environmental constraints if they were defined and active
    print("\nShadow Prices for Environmental Constraints (Part c):")
    try:
        # This will work if water_limit_constr was defined and the model solved
        # Gurobi duals for <= constraints are typically non-positive.
        # The magnitude represents the cost increase per unit decrease in the limit.
        water_dual = water_limit_constr.Pi
        print(f"  Water Constraint Shadow Price: ${abs(water_dual):.2f} per unit
reduction")
    except NameError:
        # This happens if the variable wasn't assigned (constraint commented out)
        print("  Water Constraint Shadow Price: Not active in this run.")
    except AttributeError:
        # This might happen if the constraint was added but is not binding, or model issue
         print("  Water Constraint Shadow Price: N/A (Constraint likely inactive or model
infeasible)")

    # --- Repeat for Pollutant Constraint ---
    try:
        pollutant_dual = pollutant_limit_constr.Pi
        print(f"  Pollutant Constraint Shadow Price: ${abs(pollutant_dual):.2f} per unit
reduction")
    except NameError:
        print("  Pollutant Constraint Shadow Price: Not active in this run.")
    except AttributeError:
         print("  Pollutant Constraint Shadow Price: N/A (Constraint likely inactive or
model infeasible)")
    # --- End of added code ---

elif m.status == GRB.INFEASIBLE:
     print("\nModel is infeasible. Cannot meet all constraints.")
     print("Check the constraints, especially any added environmental limits.")

else:
    print("\nNo optimal solution found. Status code:", m.status)
```

3. Code Result:

```
Gurobi Optimizer version 12.0.1 build v12.0.1rc0 (win64 - Windows 10.0 (19045.2))

CPU model: AMD Ryzen 5 5500, instruction set [SSE2|AVX|AVX2]
Thread count: 6 physical cores, 12 logical processors, using up to 12 threads

Optimize a model with 11 rows, 9 columns and 18 nonzeros
Model fingerprint: 0xe6baea44
Coefficient statistics:
  Matrix range     [1e+00, 1e+00]
  Objective range  [2e+02, 5e+03]
  Bounds range     [0e+00, 0e+00]
  RHS range        [5e+01, 2e+02]
Presolve removed 8 rows and 4 columns
Presolve time: 0.00s
Presolved: 3 rows, 5 columns, 7 nonzeros

Iteration    Objective       Primal Inf.    Dual Inf.      Time
       0    4.8000000e+04   2.500000e+01   0.000000e+00      0s
       3    2.0300000e+05   0.000000e+00   0.000000e+00      0s

Solved in 3 iterations and 0.01 seconds (0.00 work units)
Optimal objective  2.030000000e+05

Optimal total cost: $203000.00

2025 production by process:
  Process 1: 120.00 kt  (cap=120)
  → Shadow price: $-100.00 per kt
  Process 2: 30.00 kt  (cap=50)

  Process 3: 0.00 kt  (cap=0)
  → Shadow price: $-200.00 per kt

2035 production by process and expansions:
  Process 1: produce 150.00 kt  /  capacity 150.00 kt  (expanded 30.00 kt)
  Process 2: produce 50.00 kt  /  capacity 50.00 kt  (expanded 0.00 kt)
  Process 3: produce 0.00 kt  /  capacity 0.00 kt  (expanded 0.00 kt)

Shadow Prices for Environmental Constraints (Part c):
  Water Constraint Shadow Price: N/A (Constraint likely inactive or model infeasible)
  Pollutant Constraint Shadow Price: N/A (Constraint likely inactive or model infeasible)
```

4. Results and Interpretation

Question 3- Part a) What is the least cost option for meeting demand in 2025 and 2035?

Based on the Gurobi output:

- **Optimal Total Cost**: The minimum total cost to meet demand in 2025 and 2035 is **$203,000.00**.
- **2025 Production Plan**:
  - Process 1: Produce **120.00 kt** (Utilizing full initial capacity of 120 kt).
  - Process 2: Produce **30.00 kt** (Using 30 kt out of 50 kt initial capacity).
  - Process 3: Produce **0.00 kt** (Initial capacity is 0 kt).
  - *Interpretation*: To meet the 150 kt demand in 2025, the model prioritizes Process 1 (presumably due to lower operational cost, although the model considers the combined costs over both periods) up to its capacity, then uses Process 2 for the remaining demand. Process 3 is not used.
  - *Shadow Prices (2025 Capacity)*:
    - Process 1: -$100.00 per kt. This suggests that if the initial capacity of Process 1 were *increased* by 1 kt, the total optimal cost would *decrease* by $100. This negative value arises because the variable's upper bound (the capacity constraint) is active.
    - Process 3: -$200.00 per kt. Similarly, if Process 3 *had* 1 kt of initial capacity that could be used, the total cost would decrease by $200. This reflects the lower operating cost ($200/kt) of Process 3 compared to Processes 1 ($300/kt) and 2 ($400/kt). Process 2 capacity is not fully utilized, so its shadow price is $0 (not shown in output but implied).
- **2035 Production Plan & Expansion**:

- Process 1: Expand capacity by **30.00 kt** (new capacity = 120 + 30 = 150 kt). Produce **150.00 kt**.
- Process 2: Expand capacity by **0.00 kt** (capacity remains 50 kt). Produce **50.00 kt**.
- Process 3: Expand capacity by **0.00 kt** (capacity remains 0 kt). Produce **0.00 kt**.
- *Interpretation*: To meet the 200 kt demand in 2035, the model chooses to expand and fully utilize Process 1 and fully utilize the existing capacity of Process 2. Process 3 remains unused, likely because the combination of its operational cost ($200/kt) and high investment cost ($5000/kt/yr) makes expanding Process 1 ($300/kt operation, $3000/kt/yr investment) more economical for meeting the additional demand.

Question 3- Part b) Impact of Individual Environmental Policies (20% Reduction in 2035)

To answer this, you would need to perform the following steps for *each* policy (water, fuel, electricity, pollutants) individually:

1. **Calculate 2025 Baseline**: Determine the total consumption/emission in 2025 using the optimal production levels found in part (a) and the parameters from Table 11.
    - Water: (5 × 120) + (3 × 30) +(1 × 0) = 600 + 90 = 690 units
    - Electricity: (3 ×120) + (4 × 30) + (10 × 0) = 360 + 120 = 480 units
    - Fuel: (6 × 120) + (8 × 30) + (3 × 0) = 720 + 240 = 960 units
    - Pollutants: (10 × 120) + (2 × 30) + (5 × 0) = 1200 + 60 = 1260 units

2. **Set 2035 Target**: Calculate 80% of the 2025 baseline for the specific resource.

- Water Target: 0.80×690=552 units
- Electricity Target: 0.80×480=384 units
- Fuel Target: 0.80×960=768 units
- Pollutant Target: 0.80×1260=1008 units

3. **Add Constraint**: Modify the steel_nexus.py code to add *one* new constraint limiting the total 2035 consumption/emission to the target calculated in step 2. For example, for the water target: m.addConstr(gp.quicksum(water[p] * x35[p] for p in processes) <= 552, name="WaterLimit35")
4. **Re-solve**: Run the modified code.
5. **Analyse**: Compare the new optimal cost, x35p values, and Ep values to the base case results from part (a) to determine the impact of that specific policy. Repeat steps 3-5 for electricity, fuel, and pollutants separately.

6. **Results for Part B:**

Impact of Water Constraint (80% Reduction Target in 2035)

```
Optimize a model with 12 rows, 9 columns and 21 nonzeros
Model fingerprint: 0xd2d2366e
Coefficient statistics:
  Matrix range     [1e+00, 5e+00]
  Objective range  [2e+02, 5e+03]
  Bounds range     [0e+00, 0e+00]
  RHS range        [5e+01, 6e+02]
Presolve removed 10 rows and 5 columns
Presolve time: 0.00s
Presolved: 2 rows, 4 columns, 8 nonzeros

Iteration    Objective       Primal Inf.    Dual Inf.      Time
       0   -1.6000000e+33   5.000000e+29   1.600000e+03      0s
       3    5.6240000e+05   0.000000e+00   0.000000e+00      0s

Solved in 3 iterations and 0.01 seconds (0.00 work units)
Optimal objective  5.624000000e+05

Optimal total cost: $562400.00

2025 production by process:
  Process 1: 120.00 kt  (cap=120)
  → Shadow price: $-100.00 per kt
  Process 2: 30.00 kt  (cap=50)

  Process 3: 0.00 kt  (cap=0)
  → Shadow price: $-200.00 per kt

2035 production by process and expansions:
  Process 1: produce 56.00 kt  /  capacity 120.00 kt  (expanded 0.00 kt)
  Process 2: produce 64.00 kt  /  capacity 64.00 kt  (expanded 14.00 kt)
  Process 3: produce 80.00 kt  /  capacity 80.00 kt  (expanded 80.00 kt)
```

Comparing the new results (Water Constraint Scenario) to the Base Case scenario:

Optimal Total Cost:

Base Case Cost: $203,000.00

Water Constraint Cost: **$562,400.00**

Interpretation: Implementing the water reduction target significantly increases the total optimal cost by $359,400.00. This indicates that meeting the water constraint forces the company into a much more expensive operational and expansion strategy.

2025 Production Plan:

The 2025 production plan remains unchanged:

Process 1: 120.00 kt

Process 2: 30.00 kt

Process 3: 0.00 kt

Interpretation: This is expected, as the water constraint only applies to the year 2035.

2035 Production Plan & Expansion:

Base Case:

Process 1: Produce 150.00 kt (Expand by 30.00 kt, New Capacity 150 kt)

Process 2: Produce 50.00 kt (Expand by 0.00 kt, Capacity 50 kt)

Process 3: Produce 0.00 kt (Expand by 0.00 kt, Capacity 0 kt)

Water Constraint Scenario:

Process 1: Produce **56.00 kt** (Expand by **0.00 kt**, Capacity 120 kt)

Process 2: Produce **64.00 kt** (Expand by **14.00 kt**, New Capacity 64 kt)

Process 3: Produce **80.00 kt** (Expand by **80.00 kt**, New Capacity 80 kt)

Interpretation: The water constraint causes a dramatic shift in the 2035 strategy:

Production significantly shifts *away* from Process 1 (from 150 kt down to 56 kt) because it is the most water-intensive (5 units/kt). The previously planned expansion for Process 1 is cancelled.

Production shifts *towards* Process 2 (from 50 kt up to 64 kt) and requires a small expansion (14 kt). Process 2 is moderately water-intensive (3 units/kt).

Most notably, the company is now forced to invest heavily in and fully utilize **Process 3** (producing 80 kt, expanding by the maximum allowed 80 kt). Process 3 is the least water-intensive (1 unit/kt).

This shift occurs despite Process 3 having the highest investment cost ($5000/kt/yr) and Process 2 having a higher investment cost ($4000/kt/yr) than Process 1 ($3000/kt/yr). The model finds it cheaper overall to invest in and operate the less water-intensive processes (even with higher investment costs) than to violate the strict water limit using the originally cheaper Process 1.

Summary for Question 3b (Water Scenario):

The impact of limiting 2035 water consumption to 80% of 2025 levels is substantial:

The total cost more than doubles.

The 2035 production strategy shifts dramatically away from the water-intensive Process 1.

Significant investment and production are now required for Process 3 (the least water-intensive but most expensive to build), along with a small expansion of Process 2. Process 1 capacity becomes underutilized in 2035.

Impact of Electricity Constraint (80% Reduction Target in 2035)

```
Optimize a model with 12 rows, 9 columns and 21 nonzeros
Model fingerprint: 0x505f3f28
Coefficient statistics:
  Matrix range     [1e+00, 1e+01]
  Objective range  [2e+02, 5e+03]
  Bounds range     [0e+00, 0e+00]
  RHS range        [5e+01, 4e+02]
Presolve removed 10 rows and 4 columns
Presolve time: 0.00s
Presolved: 2 rows, 5 columns, 10 nonzeros

Iteration    Objective       Primal Inf.    Dual Inf.      Time
       0   -7.1175700e+04   1.326255e+02   0.000000e+00      0s

Solved in 1 iterations and 0.01 seconds (0.00 work units)
Infeasible model
No optimal solution found. Status code: 3
```

The Gurobi output provided clearly states:

- **Infeasible model**

- **No optimal solution found. Status code: 3** (Status code 3 typically indicates infeasibility in Gurobi)

Interpretation:

This result means that **it is impossible** to find a production plan (values for x351, x352, x353) and expansion plan (E1, E2, E3) for 2035 that satisfies *all* the model's constraints simultaneously when the electricity limit is imposed.

Specifically, there is no way to:

1. Meet the 2035 demand of 200 kt (∑x35p≥200).
2. Respect the capacity constraints for each process (x35p ≤ cap0p + Ep and Ep≤ deltap).
3. **AND** keep the total 2035 electricity consumption at or below the target of 384 units (∑electricity[p] · x35p ≤ 384).

Reasoning for Infeasibility:

- The electricity consumption rates are: Process 1: 3 units/kt, Process 2: 4 units/kt, Process 3: 10 units/kt.
- Notice that Process 3, which was essential for meeting demand under the strict *water* constraint due to its low water usage, is extremely electricity-intensive (10 units/kt).
- To meet the 200 kt demand in 2035, the company must produce a certain amount using a combination of the three processes within their potential expanded capacities (Process 1: max 220 kt, Process 2: max 150 kt, Process 3: max 80 kt).
- It appears that any combination of processes capable of producing 200 kt/year inherently requires more than 384 units of electricity. The processes with lower electricity consumption (P1, P2) might not have enough combined *potential* capacity, or using the high-electricity Process 3 (even minimally) pushes the total consumption over the very strict limit of 384 units.

Summary for Question 3b (Electricity Scenario):

The impact of limiting 2035 electricity consumption to 80% of 2025 levels is that **the target is unattainable**. It makes the production problem infeasible, meaning the company cannot meet its 2035 demand under this environmental restriction with the available processes and expansion limits.

Impact of Fuel Constraint (80% Reduction Target in 2035)

```
Optimize a model with 12 rows, 9 columns and 21 nonzeros
Model fingerprint: 0x5174ed32
Coefficient statistics:
  Matrix range     [1e+00, 8e+00]
  Objective range  [2e+02, 5e+03]
  Bounds range     [0e+00, 0e+00]
  RHS range        [5e+01, 8e+02]
Presolve removed 9 rows and 4 columns
Presolve time: 0.01s
Presolved: 3 rows, 5 columns, 10 nonzeros

Iteration    Objective       Primal Inf.    Dual Inf.      Time
       0    8.9247000e+03   8.256275e+01   0.000000e+00      0s

Solved in 2 iterations and 0.01 seconds (0.00 work units)
Infeasible model
No optimal solution found. Status code: 3
```

Similar to the electricity scenario, the Gurobi output you provided clearly states:

- **Infeasible model**
- **No optimal solution found. Status code: 3**

Interpretation:

This result means that **it is impossible** to find a production plan (x35p) and expansion plan (Ep) for 2035 that satisfies *all* the model's constraints simultaneously when this fuel limit is imposed.

Specifically, there is no way to:

1. Meet the 2035 demand of 200 kt (∑x35p ≥ 200).

2. Respect the capacity constraints for each process ($x35p \leq cap0p + Ep$ and $Ep \leq deltap$).
3. **AND** keep the total 2035 fuel consumption at or below the target of 768 units ($\sum fuel[p] \cdot x35p \leq 768$).

Reasoning for Infeasibility:

- The fuel consumption rates are: Process 1: 6 units/kt, Process 2: 8 units/kt, Process 3: 3 units/kt.
- While Process 3 is the most fuel-efficient, Process 2 is the most fuel-intensive, and Process 1 also uses a significant amount of fuel.
- To meet the 200 kt demand in 2035, the company must produce using a combination of the three processes within their potential expanded capacities.
- It appears that any combination of processes capable of producing the required 200 kt/year inherently requires more than 768 units of fuel. The relatively high fuel consumption of Processes 1 and 2 makes it difficult to meet demand without exceeding the strict fuel limit of 768 units, even with the contribution of the fuel-efficient Process 3.

Summary for Question 3b (Fuel Scenario):

The impact of limiting 2035 fuel consumption to 80% of 2025 levels is that **the target is unattainable**. It makes the production problem infeasible, meaning the company cannot meet its 2035 demand under this environmental restriction with the available processes and expansion limits.

Sources and related content

Impact of Pollutant Constraint (80% Reduction Target in 2035)

```
Optimize a model with 12 rows, 9 columns and 21 nonzeros
Model fingerprint: 0x916079ba
Coefficient statistics:
  Matrix range     [1e+00, 1e+01]
  Objective range  [2e+02, 5e+03]
  Bounds range     [0e+00, 0e+00]
  RHS range        [5e+01, 1e+03]
Presolve removed 10 rows and 5 columns
Presolve time: 0.00s
Presolved: 2 rows, 4 columns, 8 nonzeros

Iteration    Objective       Primal Inf.    Dual Inf.      Time
       0   -1.6000000e+33   1.000000e+30   1.600000e+03      0s
       3    4.1640000e+05   0.000000e+00   0.000000e+00      0s

Solved in 3 iterations and 0.01 seconds (0.00 work units)
Optimal objective  4.164000000e+05

Optimal total cost: $416400.00

2025 production by process:
  Process 1: 120.00 kt  (cap=120)
  → Shadow price: $-100.00 per kt
  Process 2: 30.00 kt  (cap=50)

  Process 3: 0.00 kt  (cap=0)
  → Shadow price: $-200.00 per kt

2035 production by process and expansions:
  Process 1: produce 76.00 kt  /  capacity 120.00 kt  (expanded 0.00 kt)
  Process 2: produce 124.00 kt  /  capacity 124.00 kt  (expanded 74.00 kt)
  Process 3: produce 0.00 kt  /  capacity 0.00 kt  (expanded 0.00 kt)
```

Comparing the new results (Pollutant Constraint Scenario) to the Base Case scenario:

1. **Optimal Total Cost**:
    - Base Case Cost: $203,000.00
    - Pollutant Constraint Cost: **$416,400.00**
    - *Interpretation*: Implementing the pollutant reduction target significantly increases the total optimal cost by $213,400.00. While this is a substantial

increase, it's less drastic than the cost increase caused by the water constraint. This suggests meeting the pollutant target is expensive but less disruptive to the overall cost structure than meeting the water target.

2. **2025 Production Plan**:
    - The 2025 production plan remains unchanged:
        - Process 1: 120.00 kt
        - Process 2: 30.00 kt
        - Process 3: 0.00 kt
    - *Interpretation*: As expected, the 2035 constraint doesn't alter the optimal 2025 plan.
3. **2035 Production Plan & Expansion**:
    - **Base Case**:
        - Process 1: Produce 150.00 kt (Expand by 30.00 kt, New Capacity 150 kt)
        - Process 2: Produce 50.00 kt (Expand by 0.00 kt, Capacity 50 kt)
        - Process 3: Produce 0.00 kt (Expand by 0.00 kt, Capacity 0 kt)
    - **Pollutant Constraint Scenario**:
        - Process 1: Produce **76.00 kt** (Expand by **0.00 kt**, Capacity 120 kt)
        - Process 2: Produce **124.00 kt** (Expand by **74.00 kt**, New Capacity 124 kt)
        - Process 3: Produce **0.00 kt** (Expand by **0.00 kt**, Capacity 0 kt)
    - *Interpretation*: The pollutant constraint causes a significant shift in the 2035 strategy, different from the water constraint scenario:
        - Production significantly shifts *away* from Process 1 (from 150 kt down to 76 kt) because it is the highest pollutant emitter (10 units/kt). The planned expansion for Process 1 is cancelled.

- Production shifts heavily *towards* Process 2 (from 50 kt up to 124 kt), requiring a substantial expansion (74.00 kt). Process 2 is the least polluting process (2 units/kt).
- Process 3 remains unused (0 kt production, 0 kt expansion). Although Process 3 has moderate pollution (5 units/kt), its high investment cost likely makes it less favorable than expanding the very low-pollution Process 2 to meet the constraint.
- The total 2035 production is 76 + 124 = 200 kt, meeting the demand.

Summary for Question 3b (Pollutant Scenario):

The impact of limiting 2035 pollutant emissions to 80% of 2025 levels is:

- The total cost increases significantly, but less than under the water constraint.
- The 2035 production strategy shifts away from the high-polluting Process 1.
- Significant investment and production are now required for Process 2 (the least polluting process). Process 3 remains unused. Process 1 capacity becomes underutilized in 2035.

Question 3- Part c) What would be the estimated cost per water ($/unit) and cost per pollutant ($/unit) to the company?

Estimated cost per water ($/unit):

```
Optimize a model with 12 rows, 9 columns and 21 nonzeros
Model fingerprint: 0xd2d2366e
Coefficient statistics:
  Matrix range     [1e+00, 5e+00]
  Objective range  [2e+02, 5e+03]
  Bounds range     [0e+00, 0e+00]
  RHS range        [5e+01, 6e+02]
Presolve removed 10 rows and 5 columns
Presolve time: 0.01s
Presolved: 2 rows, 4 columns, 8 nonzeros

Iteration    Objective       Primal Inf.    Dual Inf.      Time
       0   -1.6000000e+33   5.000000e+29   1.600000e+03      0s
       3    5.6240000e+05   0.000000e+00   0.000000e+00      0s

Solved in 3 iterations and 0.01 seconds (0.00 work units)
Optimal objective  5.624000000e+05

Optimal total cost: $562400.00

2025 production by process:
  Process 1: 120.00 kt  (cap=120)
  → Shadow price: $-100.00 per kt
  Process 2: 30.00 kt  (cap=50)

  Process 3: 0.00 kt  (cap=0)
  → Shadow price: $-200.00 per kt

2035 production by process and expansions:
  Process 1: produce 56.00 kt  /  capacity 120.00 kt  (expanded 0.00 kt)
  Process 2: produce 64.00 kt  /  capacity 64.00 kt  (expanded 14.00 kt)
  Process 3: produce 80.00 kt  /  capacity 80.00 kt  (expanded 80.00 kt)

Shadow Prices for Environmental Constraints (Part c):
  Water Constraint Shadow Price: $2050.00 per unit reduction
  Pollutant Constraint Shadow Price: Not active in this run.
```

The output clearly states: Water Constraint Shadow Price: $2050.00 per unit reduction.

Interpretation: This means the estimated marginal cost to the company for reducing water consumption is $2050.00 per unit of water, when forced to meet the 80% reduction target compared to the 2025 baseline. In other words, if the policy required reducing water consumption by one *more* unit (below the 552 unit target), the total operational and investment cost would increase by approximately $2050.

Estimated cost per pollutant ($/unit):

```
Optimize a model with 12 rows, 9 columns and 21 nonzeros
Model fingerprint: 0x916079ba
Coefficient statistics:
  Matrix range     [1e+00, 1e+01]
  Objective range  [2e+02, 5e+03]
  Bounds range     [0e+00, 0e+00]
  RHS range        [5e+01, 1e+03]
Presolve removed 10 rows and 5 columns
Presolve time: 0.01s
Presolved: 2 rows, 4 columns, 8 nonzeros

Iteration    Objective       Primal Inf.    Dual Inf.      Time
       0   -1.6000000e+33   1.000000e+30   1.600000e+03      0s
       3    4.1640000e+05   0.000000e+00   0.000000e+00      0s

Solved in 3 iterations and 0.01 seconds (0.00 work units)
Optimal objective  4.164000000e+05

Optimal total cost: $416400.00

2025 production by process:
  Process 1: 120.00 kt  (cap=120)
  → Shadow price: $-100.00 per kt
  Process 2: 30.00 kt  (cap=50)

  Process 3: 0.00 kt  (cap=0)
  → Shadow price: $-200.00 per kt

2035 production by process and expansions:
  Process 1: produce 76.00 kt  /  capacity 120.00 kt  (expanded 0.00 kt)
  Process 2: produce 124.00 kt  /  capacity 124.00 kt  (expanded 74.00 kt)
  Process 3: produce 0.00 kt  /  capacity 0.00 kt  (expanded 0.00 kt)

Shadow Prices for Environmental Constraints (Part c):
  Water Constraint Shadow Price: N/A (Constraint likely inactive or model infeasible)
  Pollutant Constraint Shadow Price: $512.50 per unit reduction
```

The output shows: Pollutant Constraint Shadow Price: $512.50 per unit reduction

Interpretation: This means the estimated marginal cost to the company for reducing pollutant emissions is **$512.50 per unit** of pollutant, when forced to meet the 80% reduction target compared to the 2025 baseline. So, if the policy required reducing pollutants by one *more* unit (below the 1008 unit target), the total operational and investment cost would increase by approximately $512.50.

Question 3- Part d) What would the production and capacity change be when accounting for both the water and pollution targets?

```
Optimize a model with 13 rows, 9 columns and 24 nonzeros
Model fingerprint: 0x018dbaf4
Coefficient statistics:
  Matrix range     [1e+00, 1e+01]
  Objective range  [2e+02, 5e+03]
  Bounds range     [0e+00, 0e+00]
  RHS range        [5e+01, 1e+03]
Presolve removed 10 rows and 5 columns
Presolve time: 0.00s
Presolved: 3 rows, 4 columns, 12 nonzeros

Iteration    Objective       Primal Inf.    Dual Inf.      Time
       0   -1.6000000e+33   1.000000e+30   1.600000e+03      0s
       4    5.8640000e+05   0.000000e+00   0.000000e+00      0s

Solved in 4 iterations and 0.01 seconds (0.00 work units)
Optimal objective  5.864000000e+05

Optimal total cost: $586400.00

2025 production by process:
  Process 1: 120.00 kt  (cap=120)
  → Shadow price: $-100.00 per kt
  Process 2: 30.00 kt  (cap=50)

  Process 3: 0.00 kt  (cap=0)
  → Shadow price: $-200.00 per kt

2035 production by process and expansions:
  Process 1: produce 48.73 kt  /  capacity 120.00 kt  (expanded 0.00 kt)
  Process 2: produce 78.55 kt  /  capacity 78.55 kt  (expanded 28.55 kt)
  Process 3: produce 72.73 kt  /  capacity 72.73 kt  (expanded 72.73 kt)

Shadow Prices for Environmental Constraints (Part c):
  Water Constraint Shadow Price: $850.00 per unit reduction
  Pollutant Constraint Shadow Price: $300.00 per unit reduction
```

Comparing the Part (d) results to the Base Case (Part a) results for 2035:

1. Optimal Total Cost:
    - Base Case Cost: $203,000.00
    - Combined Constraint Cost: $586,400.00
    - *Change*: The total cost increases substantially by $383,400.00 when both constraints are active. This cost is higher than when only the pollutant constraint was active ($416,400) and slightly higher than when only the

water constraint was active ($562,400), indicating the combined restrictions force further, more expensive adjustments.

2. Process 1:
    - Base Case: Produce 150.00 kt / Expand 30.00 kt
    - Combined Constraints: Produce 48.73 kt / Expand 0.00 kt
    - *Change*: Production is drastically reduced (by 101.27 kt), and the planned expansion is cancelled. The initial capacity (120 kt) is now heavily underutilized. This is driven by Process 1's high water usage (5 units/kt) and high pollutant emission (10 units/kt).
3. Process 2:
    - Base Case: Produce 50.00 kt / Expand 0.00 kt
    - Combined Constraints: Produce 78.55 kt / Expand 28.55 kt
    - *Change*: Production increases significantly (by 28.55 kt), requiring a new capacity expansion of 28.55 kt. This process becomes more important due to its low pollutant emission (2 units/kt) and moderate water use (3 units/kt).
4. Process 3:
    - Base Case: Produce 0.00 kt / Expand 0.00 kt
    - Combined Constraints: Produce 72.73 kt / Expand 72.73 kt
    - *Change*: This process, previously unused, is now heavily utilized, requiring a large expansion (72.73 kt out of a maximum possible 80 kt). This is driven by its very low water usage (1 unit/kt) and moderate pollutant emission (5 units/kt), despite its high investment cost.

Summary of Changes for Part (d):

When both water and pollutant reduction targets are enforced simultaneously for 2035:

- The company must significantly shift production away from Process 1.
- It needs to invest in expanding and utilizing both Process 2 (low pollution) and Process 3 (low water use).

- The total production (48.73 + 78.55 + 72.73 = 200 kt) still meets the 2035 demand.
- The overall cost is significantly higher than the base case and reflects the combined difficulty of meeting both environmental restrictions simultaneously.

Question 2: Manufacturing Profit Maximization

1. Problem Description & Setup

A manufacturing company aims to maximize its monthly profit by deciding how many units of three different products (Product 1, Product 2, Product 3) to produce. Production requires three different parts (Part 1, Part 2, Part 3), which can be sourced from three different suppliers (Supplier 1, Supplier 2, Supplier 3).

The problem involves considering:

- **Part Requirements**: Each product requires a specific number of each part (Table 8).
- **Supplier Constraints**: Each supplier has a monthly limit on the quantity of each part they can provide and charges a specific price per part (Table 9).
- **Production Constraints**: Each product has a minimum and maximum monthly production level, a selling price, and a production cost (excluding part costs) (Table 10).
- **Objective:** Determine the optimal production quantity for each product and the optimal purchase quantity of each part from each supplier to maximize the total monthly profit (Total Revenue - Total Production Cost - Total Part Purchase Cost).

Additional scenarios explore the impact of:

- A maximum spending limit per supplier (Part d & e).
- Introducing a new international supplier (Part f).

2. Mathematical Model (Baseline)

Decision Variables:

- Makep: Number of units of product p to produce per month ($p \in 1,2,3$).

- $Buy_{s,n}$: Number of units of part n to buy from supplier s per month ($s \in 1,2,3$, $n \in 1,2,3$).

Parameters (from Tables 8, 9, 10):

- $needs_{p,n}$: Units of part n required for one unit of product p.
- $supplierlimits_{s,n}$: Maximum units of part n available from supplier s.
- $partcosts_{s,n}$: Cost per unit of part n from supplier s.
- $prodprice_p$: Selling price per unit of product p.
- $prodcost_p$: Production cost per unit of product p (excluding parts).
- $minprod_p$: Minimum production units for product p.
- $maxprod_p$: Maximum production units for product p.

Objective Function: Maximize Total Profit Z:

$$Z=\sum_p(prodprice_p-prodcost_p)\cdot Make_p-\sum_s\sum_n partcosts_{s,n}\cdot Buy_{s,n}$$

Constraints:

1. Part Balance: For each part n, the total units bought must be at least the total units needed for production.
   $\sum_s Buy_{s,n} \geq \sum_p needs_{p,n}\cdot Make_p \quad \forall n \in 1,2,3$
2. Supplier Limits: The quantity bought from a supplier for a specific part cannot exceed their limit.
   $Buy_{s,n} \leq supplierlimits_{s,n} \quad \forall s \in 1,2,3, n \in 1,2,3$
3. Production Limits: Production for each product must be within its minimum and maximum bounds.
   $minprod_p \leq Make_p \leq maxprod_p \quad \forall p \in 1,2,3$

4. Non-negativity: All decision variables must be non-negative. Makep≥0,Buys,n≥0

(Note: Constraints for Parts d and f involve modifications to this baseline model as described later.)

3. Code

```
import gurobipy as gp
from gurobipy import GRB

# --- Data ---
products = [1, 2, 3]
parts = [1, 2, 3]
# Original:
suppliers = [1, 2, 3]
# Modified for Part f:
#suppliers = [1, 2, 3, 4]

# Parts needed per product (Table 8)
# Using (Product, Part): Need format
needs = {
    (1, 1): 25, (1, 2): 10, (1, 3): 15,
    (2, 1): 12, (2, 2): 7,  (2, 3): 2,
    (3, 1): 5,  (3, 2): 4,  (3, 3): 10
}

# Supplier limits (Table 9 + Part f)
# Using (Supplier, Part): Value format
supplier_limits = {
    (1, 1): 1000, (2, 1): 1000, (3, 1): 1000, # Original Part 1 limits
    (1, 2): 1000, (2, 2): 1000, (3, 2): 1000, # Original Part 2 limits
    (1, 3): 1000, (2, 3): 1000, (3, 3): 1000, # Original Part 3 limits
    #(4, 1): 500,                              # New Supplier 4 Part 1 limit [cite: 25]
    #(4, 2): 500,                              # New Supplier 4 Part 2 limit [cite: 25]
    #(4, 3): 500                               # New Supplier 4 Part 3 limit [cite: 25]
}
# --- End of addition ---

# Supplier costs (Table 9 + Part f)
```

```
# Using (Supplier, Part): Value format
part_costs = {
    (1, 1): 7,  (2, 1): 9,  (3, 1): 5,      # Original Part 1 costs [cite: 15]
    (1, 2): 14, (2, 2): 12, (3, 2): 18,     # Original Part 2 costs [cite: 15]
    (1, 3): 8,  (2, 3): 9,  (3, 3): 10,     # Original Part 3 costs [cite: 15]
    #(4, 1): 6,                              # New Supplier 4 Part 1 cost [cite: 25]
    #(4, 2): 13,                             # New Supplier 4 Part 2 cost [cite: 25]
    #(4, 3): 9                               # New Supplier 4 Part 3 cost [cite: 25]
}

# Product information (Table 10 - Using User's Corrected Interpretation)
prod_price = {1: 800, 2: 700, 3: 950}
# **Corrected Production Cost for P3**
prod_cost = {1: 125, 2: 250, 3: 450}
# **Corrected Min/Max Production for P3**
min_prod = {1: 40, 2: 110, 3: 60}
max_prod = {1: 250, 2: 400, 3: 130}

# --- Model ---
m = gp.Model("Manufacturing_Profit")

# --- Decision Variables ---
# How many units of each product to make
Make = m.addVars(products, name="Make", lb=0) # Using descriptive name
# How many units of each part to buy from each supplier
Buy = m.addVars(suppliers, parts, name="Buy", lb=0) # Using descriptive name

# --- Objective Function ---
# Profit = Revenue - Production_Cost - Part_Purchase_Cost
Revenue = gp.quicksum(prod_price[p] * Make[p] for p in products)
Production_Cost = gp.quicksum(prod_cost[p] * Make[p] for p in products)
Part_Purchase_Cost = gp.quicksum(part_costs[s, n] * Buy[s, n] for s in suppliers for n in
parts)

m.setObjective(Revenue - Production_Cost - Part_Purchase_Cost, GRB.MAXIMIZE)

# --- Constraints ---

# 1. Part Balance: Total parts bought >= Total parts needed
part_balance_constrs = {}
for n in parts:
    parts_needed = gp.quicksum(needs.get((p, n), 0) * Make[p] for p in products) # Use
.get for safety
    parts_bought = gp.quicksum(Buy[s, n] for s in suppliers)
    part_balance_constrs[n] = m.addConstr(parts_bought >= parts_needed,
name=f"PartBalance_{n}")

# 2. Supplier Limits: Buy[s,n] <= Limit[s,n]
```

```
supplier_limit_constrs = {}
for s in suppliers:
    for n in parts:
         supplier_limit_constrs[s,n] = m.addConstr(Buy[s, n] <= supplier_limits.get((s,
n), 0), name=f"SupplierLimit_{s}_{n}") # Use .get

# 3. Production Limits: MinProd <= Make <= MaxProd
min_prod_constrs = {}
max_prod_constrs = {}
for p in products:
    min_prod_constrs[p] = m.addConstr(Make[p] >= min_prod[p], name=f"MinProd_{p}")
    max_prod_constrs[p] = m.addConstr(Make[p] <= max_prod[p], name=f"MaxProd_{p}")

# 4. Max Spending per Supplier (Part D)
#max_spending = 25000
#supplier_spending_constrs = {}
#for s in suppliers:
    #supplier_spending_constrs[s] = m.addConstr(
       # gp.quicksum(part_costs[s, n] * Buy[s, n] for n in parts) <= max_spending,
       # name=f"MaxSpend_{s}"
    #)

# --- Solve ---
m.optimize()

# --- Display Results (Parts a, b, c) ---
if m.status == GRB.OPTIMAL:
    print("\nOptimal Profit Level (Part b): $%.2f\n" % m.objVal)
    print("Optimal Production Levels (Part a):")
    for p in products:
        # Only print if value is significantly greater than zero
        if Make[p].x > 1e-6:
             print(f"  Product {p}: {Make[p].x:.2f} units")

    print("\nParts Bought from Suppliers (Part c):")
    # Group by supplier for readability
    for s in suppliers:
        supplier_purchases = False
        for n in parts:
            if Buy[s, n].x > 1e-6:
                if not supplier_purchases:
                    print(f"  Supplier {s}:")
                    supplier_purchases = True
                print(f"    Part {n}: {Buy[s, n].x:.2f} units")
        if not supplier_purchases:
             print(f"  Supplier {s}: (No parts bought)")

            # --- Display Sensitivity Analysis (Part e) ---
```

```
if m.status == GRB.OPTIMAL:
    print("\nOptimal Profit Level: $%.2f\n" % m.objVal)

    print("Optimal Production Levels:")
    # ... (your code to print Make[p].x values) ...

    print("\nParts Bought from Suppliers:")
    # ... (your code to print Buy[s, n].x values) ...

    # --- ADD THE SENSITIVITY ANALYSIS CODE HERE (FOR PART E) ---
   # print("\nShadow Prices (Supplier Spending Constraints - Part e):")
    # Check Supplier 1
    #s1_constr = supplier_spending_constrs[1]
    #print(f"  Supplier 1 ({s1_constr.ConstrName}): Shadow Price = {s1_constr.Pi:.4f}")

    # Check Supplier 2
    #s2_constr = supplier_spending_constrs[2]
    #print(f"  Supplier 2 ({s2_constr.ConstrName}): Shadow Price = {s2_constr.Pi:.4f}")

    # Check Supplier 3
    #s3_constr = supplier_spending_constrs[3]
    #print(f"  Supplier 3 ({s3_constr.ConstrName}): Shadow Price = {s3_constr.Pi:.4f}")
    # --- END OF SENSITIVITY ANALYSIS CODE ---

elif m.status == GRB.INFEASIBLE:
    print("\nModel is infeasible. Cannot meet all constraints.")
    # ... (optional IIS computation) ...

else:
    print("\nNo optimal solution found. Status code:", m.status)
```

4. Results and Interpretation

Part a) Optimal Production Levels (Baseline)

- **Question**: In order to maximize their profit, what is the best production level for each product?
- **Results:**
  - Product 1: 40.00 units
  - Product 2: 112.50 units

- Product 3: 130.00 units
- **Interpretation**: To maximize profit under the initial conditions, the company should produce the minimum required amount of Product 1 (40 units) and the maximum allowed amount of Product 3 (130 units). Production for Product 2 is set at 112.50 units, which falls between its minimum (110) and maximum (400) limits. This indicates that the profitability and part requirements of Products 1 and 3 push their production to their respective limits, while Product 2's level is determined by the remaining resource availability and its own profitability.

Part b) Optimal Profit Level (Baseline)

- Question: What is the optimal profit level?
- Result: $81,470.00
- Interpretation: The maximum achievable monthly profit, given the baseline constraints and costs, is $81,470.00. This figure represents the total revenue from selling the optimal quantities of products minus the production costs and the cost of purchasing the required parts from suppliers.

Part c) Parts Bought from Suppliers (Baseline)

- **Question**: How much of each part is being bought from each supplier?
- **Results:**
- **Supplier 1:** Part 1: 1000.00 units, Part 2: 707.50 units, Part 3: 1000.00 units
- **Supplier 2:** Part 1: 1000.00 units, Part 2: 1000.00 units, Part 3: 1000.00 units
- **Supplier 3:** Part 1: 1000.00 units, Part 3: 125.00 units (No Part 2 bought)
- **Interpretation:** The purchasing plan utilizes suppliers based primarily on cost, up to their supply limits. Supplier 3 provides the cheapest Part 1 ($5), Supplier 2 the cheapest Part 2 ($12), and Supplier 1 the cheapest Part 3 ($8). The model buys the maximum 1000 units of Part 1 from all suppliers, indicating high demand or strategic purchasing from the cheapest (S3) first. For Part 2, the cheapest (S2) is maxed out, and the remainder comes from S1. For Part 3, the cheapest (S1)

and second cheapest (S2) are maxed out, with the small remaining need fulfilled by S3. Supplier 3's Part 2 is the most expensive ($18) and is not utilized at all. Many suppliers are at their maximum limit (1000 units) for specific parts.

Part d) Impact of $25,000 Spending Limit per Supplier

- **Question:** How would a maximum spending limit of $25,000 per supplier change the production profile and the supply profile of each part?
- **Analysis:** A constraint was added to the model limiting the total cost of parts purchased from each supplier (s) to $25,000: ∑npartcostss,n·Buys,n≤25000.
- **Results:**
  - Optimal Profit: Decreased to $80,914.44 (a loss of $555.56).
  - Production Levels: Remained unchanged (P1: 40, P2: 112.50, P3: 130).
  - Parts Bought:
    - Supplier 1: Unchanged (Spending: $24,905).
    - Supplier 2: Part 3 purchases reduced from 1000 to 444.44 units. P1 and P2 purchases remain 1000. (Spending hits limit: $25,000).
    - Supplier 3: Part 3 purchases increased from 125 to 680.56 units. P1 purchases remain 1000. (Spending: $11,805.6).
- **Interpretation:** The spending limit directly constrains Supplier 2, who was originally spending $30,000. The model adapts by reducing purchases of the relatively cheaper Part 3 ($9) from Supplier 2 and compensating by buying more of the more expensive Part 3 ($10) from Supplier 3. This forced shift to a more expensive sourcing option causes the overall profit to decrease. The production levels did not change, indicating the primary impact was on the sourcing strategy rather than the quantity produced.

Part e) Sensitivity Analysis of Spending Limit

- **Question:** What is the impact (cost) of the $25,000 spending limitation for negotiation purposes? (Interpreted via shadow prices).

- **Analysis:** The shadow prices (dual values, .Pi attribute in Gurobi) of the spending limit constraints added in Part d were examined.
- **Results:**
  - Supplier 1 Shadow Price: $0.0000
  - Supplier 2 Shadow Price: $0.1111
  - Supplier 3 Shadow Price: $0.0000
- **Interpretation:** The shadow price indicates the change in the objective function (profit) if the constraint's right-hand side (the $25,000 limit) is relaxed by one unit.
  - For Supplier 2, the shadow price of $0.1111 means that if the spending limit for Supplier 2 were increased by $1 (to $25,001), the total profit would increase by $0.1111. This quantifies the marginal cost of the restriction for Supplier 2; every dollar the company is prevented from spending optimally with Supplier 2 (due to the cap) costs the company approx. 11 cents in lost profit.
  - The zero shadow prices for Suppliers 1 and 3 indicate their spending limits are not binding; relaxing their limits would not immediately increase profit. This information is crucial for negotiating with the regulator, highlighting that the restriction on Supplier 2 is the one causing a direct financial loss.

Part f) Impact of New International Supplier

- **Question:** How would adding a new international supplier (Supplier 4) with costs $6 (P1), $13 (P2), $9 (P3) and limits of 500 units/part change the original solution?
- **Analysis:** The baseline model (from parts a, b, c) was modified to include Supplier 4 with its associated costs and limits.
- **Results:**
  - Optimal Profit: Increased significantly to $93,011.67 (an increase of $11,541.67 from baseline).

- Production Levels: Product 1: 40.00 (unchanged), Product 2: Increased to 154.17 units, Product 3: 130.00 (unchanged).
- Parts Bought:
  - Supplier 1: P1: 1000, P2: reduced to 499.17, P3: 1000.
  - Supplier 2: P1: 1000, P2: 1000, P3: reduced to 708.33.
  - Supplier 3: P1: 1000, P2: 0, P3: reduced to 0.
  - Supplier 4: P1: 500 (max), P2: 500 (max), P3: 500 (max).
- **Interpretation:** Introducing Supplier 4 is highly advantageous. The company fully utilizes the new supplier's capacity for all parts due to its competitive pricing (cheaper P1 than S1/S2, cheaper P2 than S1, cheaper P3 than S3). This significantly reduces overall part costs, leading to a substantial profit increase. The lower part costs also make it optimal to increase production of Product 2. Purchases from more expensive sources (like S3 for Part 3, S1 for Part 2) are reduced or eliminated. This strongly suggests the company should pursue the partnership with the international supplier.

Q1:

The objective of this problem is to evaluate the optimal structure of production and trade across three countries—A, B, and C. Each country can produce using three types of raw resources: Ore, Oil, and Land. These are transformed into intermediary products: Metal, Crop, and Plastic, which in turn are used to produce final products: Machines and Food.

The model minimizes the **total economic cost**, which includes:

- Production cost for resources and products,
- Trade cost for importing/exporting between countries.

Each country has specific limitations:

- Availability of raw resources,
- Production and export capacities,
- Technical production coefficients for all product layers,
- National demand for final goods (Machines and Food).

Constraints:

For each country:

- Ore ≤ ProdLimit_Ore
- Land ≤ ProdLimit_Land
- Oil ≤ ProdLimit_Oil

Intermediary Production:

- 1 unit of Metal = c_ore × Ore + c_oil × Oil
- 1 unit of Crop = c_land × Land + c_oil × Oil
- 1 unit of Plastic = [varies per country] (e.g., Oil only or Oil + Land)

Final Product Formulas:

- Machine = c_metal × Metal + c_plastic × Plastic
- Food = c_crop × Crop + c_plastic × Plastic

Production Limits:

- Metal, Crop, Plastic, Machine, Food ≤ ProdLimit

Export Capacity:

- Trade (Sold) ≤ ExportLimit

Trade Balance (for tradable products):

- Total [Product] = Produced + Trade (Bought) – Trade (Sold)

Demand Constraints:

- Total Machine ≥ Demand_Machine
- Total Food ≥ Demand_Food

Math formula

Constrains:

For each country

Ore <= ProdLimit_Ore

Land <= ProdLimit_Land

Oil <= ProdLimit_Oil

1 unit of Metal = c_ ore * Ore + c_oil * Oil

1 unit of Crop = c_land * Land + c_oil * Oil

1 unit of plastic = 1 unit of Oil

Machine = c_metal * Metal + c_plastic * Plastic

Food = c_crop * Crop + c_plastic * plastic

Meta, Crop, Plastic, Machine, Food <= ProdLimit

For tradable products:

Total [Product] = Produced + Trade (Bought) – Trade (Sold)

Total Machine >= Demand_machine

Total Food >= Deman_Food